Venkata Lakshmi Parimala Pasupuleti

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SUMMARY

Software engineer with 5+ years of experience building and maintaining production applications in regulated housing environments. Background in full-stack development (Python, Java, React, TypeScript, PostgreSQL/Oracle) with a focus on data accuracy, compliance workflows, and application quality. Experience with housing management software at scale -- lease management, eligibility calculations, HUD compliance, and resident data workflows across a 100K+ MAU platform. Currently at Intuit integrating AI tools into production financial services. Responsible daily user of AI-assisted development tools (Claude Code). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Python / Java / C# adjacent | JavaScript / TypeScript | React / Angular adjacent | PostgreSQL / Oracle / SQL Server | REST APIs | Unit Testing / TDD | CI/CD (GitHub Actions) | Git / Version Control | AWS (S3, Lambda, RDS) | Docker | Agile / Scrum | Multi-Layer Application Architecture | Housing Compliance (HUD / LIHTC) | AI-Assisted Development (Claude Code) | Debugging & Troubleshooting

EXPERIENCE

Intuit

Jun 2026 -- Present

Software Engineer -- Virginia, USA (Hybrid)

- Build and maintain Python/FastAPI backend services and React/TypeScript frontends on AWS, shipped through GitHub Actions CI/CD with Docker.
- Integrate AI capabilities (Anthropic Claude) into financial data workflows; implement input validation, guardrails, and eval harnesses to ensure accuracy and security before release.
- Write unit tests and validation suites for LLM outputs and ML pipelines; instrument services with Prometheus for production observability.
- Responsibly use AI-assisted development tools (Claude Code) daily, reviewing and testing all generated code before shipping.

Entrata -- Residential & Affordable Housing SaaS

Oct 2020 -- Jul 2024

Senior Software Engineer -- Multi-tenant platform, 100K+ monthly active users

- Developed and maintained software applications supporting **residential housing operations** -- lease management, eligibility calculations, HUD compliance, maintenance workflows, and resident communication -- used daily by property managers across thousands of communities.
- Investigated, troubleshot, and resolved software defects across a complex multi-layer application (database, server, client); optimized 200+ SQL queries using execution plan analysis (30% performance gain).
- Implemented software enhancements and approved changes across Python, Java, PHP, React, and PostgreSQL/Oracle, with unit testing and code review on every change.
- Maintained the annual **HUD income-limit data pipeline** for 6 years -- validation, transformation, and rollback across ~180,000 domain records. Zero compliance violations.
- Built RabbitMQ-based async processing with idempotency guarantees; implemented PHI encryption (Libsodium) for resident data; led AWS S3 document migration.
- Reviewed 300+ pull requests; documented development activities and maintained runbooks for recurring operational workflows.

PROJECTS

Leasel -- Housing Compliance Application

Python · React · FastAPI · PostgreSQL · Unit Testing

Full-stack application for housing compliance operations -- document ingestion, clause extraction, compliance Q&A, and human escalation for regulated decisions. Built with TDD; automated test harness across 1,240 test cases (93.6% precision). Role-based access control throughout.

github.com/venkatalakshmiparimala/Lease_complaince_rag

AI on FHIR -- Clinical NLP Application

Python · Flask · React · TypeScript · REST APIs · Docker

RESTful API application translating clinical questions into FHIR API calls using NLP. Multi-layer architecture (database, server, client); role-based access control; containerized with Docker. Automated testing suite covering API endpoints and access boundaries.

github.com/venkatalakshmiparimala/AI_on_FHIR

AgentProbe -- API Testing & Evaluation Framework

Python · FastAPI · CI/CD · Automated Testing

15-endpoint FastAPI backend with automated regression testing via golden datasets in CI. Unit and integration test coverage across five API providers; ensures application accuracy and reliability before every release.

github.com/venkatalakshmiparimala/AgentProbe

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 | GPA 3.87

B.Tech Computer Science & Engineering — JNTU Anantapur — 2017